# A Comparative Study of Machine Learning Approaches for Postprandial Glucose Response Prediction Using CGM and Nutritional Data

**Cochiorca Oana Maria** **Lungu Laura-Vanesa**

oanamaria.cochiorca01@universitadipavia.it lauravanesa.lungu01@universitadipavia.it

## Abstract

The rise in blood glucose following a meal - its magnitude, timing, and duration - varies substantially between individuals, even in response to identical foods. This postprandial glucose response (PPGR) is a key marker of metabolic health: sustained elevations are associated with insulin resistance, pre-diabetes, and type-2 diabetes, yet reliable prediction remains an open challenge due to the combined influence of dietary, physiological, and behavioral factors. In this work, we compare different machine learning approaches for predicting PPGR using multimodal data from continuous glucose monitoring (CGM), nutritional intake, clinical measurements, and physical activity. We evaluate linear mixed-effects models, gradient boosting (XGBoost), and a tabular foundation model (TabPFN) on the CGMacros dataset using a leave-one-subject-out cross-validation framework. We investigate whether meal-specific modeling improves prediction and compare model performance across two measures of glucose response: AUC and incremental AUC. Results show that meal context plays an important role in prediction accuracy, with more standardized meals producing more reliable predictions. XGBoost and TabPFN achieve similar performance, while feature analysis highlights the importance of metabolic and meal-related characteristics in explaining glucose variability. These findings support the use of multimodal machine learning approaches for personalized PPGR prediction.

## 1 Introduction

Postprandial glucose response (PPGR) refers to the change in blood glucose levels after a meal. In healthy individuals, blood glucose typically rises after eating, reaches a peak within approximately one hour, and gradually returns to baseline over the following two to three hours.

### 1.1 Problem Domain and Clinical Motivation

Abnormal postprandial glucose levels are associated with an increased risk of metabolic disease. According to current clinical guidelines, a two-hour postprandial glucose value below 140 mg/dL is considered normal, values between 140 and 199 mg/dL indicate impaired glucose tolerance (pre-diabetes), and values of 200 mg/dL or higher are diagnostic of type-2 diabetes (T2D) (American Diabetes Association Professional Practice Committee, 2024). Furthermore, elevated postprandial glucose has been linked to cardiovascular complications and an increased likelihood of developing T2D. These associations highlight the clinical importance of understanding and predicting PPGR.

The magnitude and shape of this response curve are influenced by a combination of dietary, physiological, and behavioral factors rather than by carbohydrate intake alone. Studies have shown that meal composition, including the amount of carbohydrates, fiber, fat, and protein, affects both the magnitude and timing of the glucose response. Fiber can slow carbohydrate absorption, while fat and protein may delay gastric emptying and alter the shape of the glucose curve. In addition, factors such as physical activity, insulin sensitivity, age, body composition, and gut microbiome characteristics have also been associated with differences in PPGR. As a result, predicting glucose responses is a complex task that requires considering multiple interacting variables rather than a single nutritional measure.

Importantly, these factors do not affect all individuals in the same way. In a landmark study, Zeevi et al. (2015) demonstrated substantial inter-individual variability in glycemic responses, showing that two individuals consuming the same meal could exhibit markedly different glucose patterns. For example, foods that produced a large glucose spike in one participant sometimes generated only a modest response in another. These findings challenged the idea of universal dietary recommendations and motivated the development of personalized approaches for predicting PPGR.

### 1.2 Related Work

**Data availability and dataset limitations.**

Progress in PPGR modeling has been strongly constrained by limited data availability. Most publicly available CGM datasets focus on type 1 diabetes populations, while datasets covering healthy and type 2 diabetes individuals with detailed meal annotations remain rare. Patient-level metabolic data are also difficult to share due to privacy regulations, making publicly available multimodal resources that combine CGM recordings with macronutrient logs scarce.

**Classical machine learning approaches.**

Building on the findings of Zeevi et al. (2015), subsequent work has focused on translating that personalized prediction framework into practical and generalizable models. Mendes-Soares et al. (2019) validated a similar gradient boosting approach on an independent American cohort, confirming its applicability across populations. Since then, a range of methods including Random Forests, XGBoost, LightGBM, and Support Vector Regression have been applied to PPGR prediction due to their ability to capture non-linear feature interactions without requiring extensive feature engineering.

### 1.3 Research Question and Contributions

Despite recent progress in PPGR modeling, it remains unclear how different modeling paradigms compare under consistent evaluation settings, particularly when using modern multimodal CGM datasets. The benefit of incorporating meal-level structure - such as stratifying by meal type - has also not been systematically assessed across model families. In this work, we address these gaps using CGMacros (Das et al., 2025), a multimodal dataset containing CGM signals, meal-level nutritional intake, physical activity, and clinical metadata for 45 subjects across healthy, pre-diabetic, and type-2 diabetes cohorts. Specifically, we:

- **Compare three model families** – linear mixed-effects models, gradient boosting (XGBoost), and a tabular foundation model (TabPFN) – under a leave-one-subject-out (LOSO) cross-validation framework, on two complementary targets of postprandial glucose exposure: AUC and iAUC.
- **Evaluate the effect of meal-type stratification**, assessing whether training separate models for breakfast, lunch, and dinner improves performance over a single general model.
- **Analyze model interpretability and prediction fairness**, enabling a cross-model comparison of feature importance.

## 2 Data and Preprocessing

### 2.1 The CGMacros Dataset

CGMacros (Das et al., 2025) is a publicly available multimodal dataset collected from 45 adults at the Sansum Diabetes Research Institute over ten consecutive days. Each participant simultaneously wore two blinded continuous glucose monitors throughout the study period: an Abbott FreeStyle Libre Pro affixed to the upper arm (15-minute sampling interval) and a Dexcom G6 Pro affixed to the abdomen (5-minute sampling interval). A Fitbit Sense smartwatch was worn continuously to record heart rate (HR), minute-level energy expenditure (Calories Activity), and metabolic equivalent of task (METs) estimates. All sensor streams were aligned to a common 1-minute timeline per participant. Since CGM readings are sparser than this 1-minute grid, glucose values between sensor measurements were linearly interpolated in the dataset files.

Participants logged three meals per day using MyFitnessPal and photographed each meal before and after eating for estimating the amount of consumed calories. Importantly, the three meal types differ in how tightly macronutrients were controlled: **breakfasts** were standardized protein shakes with known compositions, **lunches** were ordered from a single restaurant chain with documented nutritional content, and **dinners** were freely chosen and self-reported – the least reliable of the three. Meals (excluding snacks) were separated by at least three hours to minimize carry-over effects between meals.

In addition, the dataset includes a fasting blood panel (HbA1c, fasting glucose, insulin, BMI, and lipid markers), gut microbiome data from Viome sequencing, and 22 composite gut health scores per participant. Based on their HbA1c values, participants were classified into three metabolic groups: **healthy** ($n = 15$, HbA1c $< 5.7\%$), **pre-diabetic** ($n = 16$, 5.7–6.4%), and **type-2 diabetic** ($n = 14$, HbA1c $> 6.4\%$).

### 2.2 CGM Device Selection

Although participants wore two CGM sensors simultaneously, all experiments in this work are based exclusively on Freestyle Libre data. The original dataset paper reports a systematic offset of up to 40 mg/dL between the two devices across all metabolic groups, with Dexcom readings consistently higher – likely due to differences in anatomical placement and subcutaneous fat thickness at the respective sensor sites. Beyond the absolute offset, the Dexcom's 5-minute sampling rate produces noisier postprandial curves compared to the Libre's smoother 15-minute trace. Figure 1 illustrates this difference across one representative day for a healthy, pre-diabetic, and T2D participant: while both devices capture the same postprandial peaks, the Dexcom trace shows consistently higher absolute values and more high-frequency noise throughout the day. Given the relatively small dataset size, the Libre was selected as the sole CGM source to avoid introducing device-related noise into the prediction targets.

### 2.3 Data Quality and Filtering

Before constructing the feature set, the raw per-subject CSV files were audited for structural inconsistencies and data quality issues. One subject was excluded entirely because the activity column was absent from their recording – indicating the Fitbit was not worn rather than a random data gap (Missing Not At Random). The METs field, while present in the majority schema, was inconsistently replaced by a categorical `Intensity` variable across eleven subjects, and was therefore excluded from all analyses. Activity was instead represented using only the `Calories (Activity)` feature, which was available for all remaining 44 subjects.

Meal events were then filtered to retain only breakfast, lunch, and dinner records, as snack entries were sparse and nutritionally heterogeneous, making them unsuitable for systematic analysis. Additional quality filters were applied to exclude logging errors and sensor artefacts: meals with zero or missing calorie entries, macronutrient values outside the valid range of 0–176 g, or glucose readings below the sensor floor of 40 mg/dL were all discarded. After all filtering steps, the final dataset comprised **1,170 meal events** from **44 subjects** (Healthy: 15, Pre-diabetic: 15, T2D: 14).

### 2.4 Feature Engineering

For each retained meal event, a postprandial glucose trace of 9 Libre readings was extracted at 15-minute intervals over a 120-minute window starting from meal onset. Two prediction targets were derived from this trace, as illustrated in Figure 1: the **total AUC**, computed via the trapezoidal rule over the raw glucose values, and the **incremental AUC (iAUC)**, which integrates only the area above the pre-meal glucose level (`Baseline Libre glucose`). While AUC captures overall glycaemic burden including the baseline level, iAUC isolates the reactive postprandial spike – the two targets are therefore complementary and both are predicted in all experiments.

Two additional features were derived from the available data. **HOMA-IR** was computed from the fasting blood panel as a proxy for insulin resistance. **Physical activity** was summarized as the total Fitbit-estimated calorie expenditure in the 60 minutes before each meal (`Activity_pre_sum`) and in the 120 minutes after meal onset (`Activity_post_sum`). Pre-meal activity captures the effect of recent exercise on insulin sensitivity, while post-meal activity reflects energy expenditure during the glucose absorption window – both of which have been shown to modulate postprandial glucose response (Bellini et al., 2024). Table 1 summarizes the complete feature set used across all experiments.

Table 1: Input features used across all experiments.

| Category | Feature |
|---|---|
| Meal composition | Carbs, Protein, Fat, Fiber (kcal)<br>Consumed Calories (kcal) |
| CGM | Baseline Libre glucose (mg/dL) |
| Clinical | Age, Gender, BMI<br>HbA1c, Fasting glucose, Insulin<br>HOMA-IR, Triglycerides<br>Cholesterol, HDL, Non-HDL,<br>LDL, VLDL, Cho/HDL ratio |
| Activity | Activity_pre_sum (kcal, 1hr)<br>Activity_post_sum(kcal, 2hr) |

### 2.5 Exploratory Data Analysis

To verify the cohort composition and motivate the modelling choices, we conducted a brief exploratory analysis of the final dataset. Figure 2 shows the distribution of key demographic and clinical features across the three metabolic groups. As expected, HbA1c and fasting glucose increase monotonically from healthy to pre-diabetic to T2D, and HOMA-IR follows the same trend, confirm-

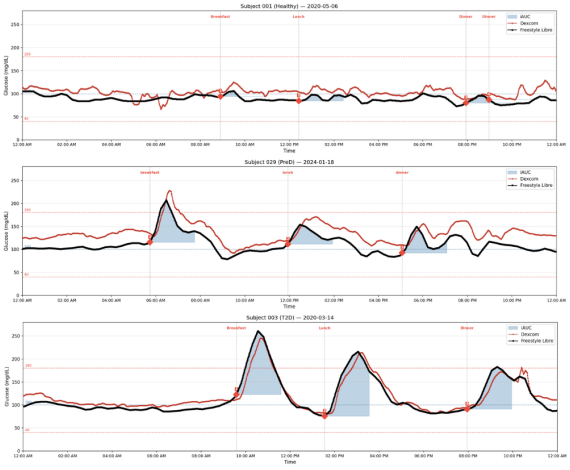


Figure 1: 24-hour CGM recordings for one representative participant from each metabolic group (Healthy, Pre-diabetic, T2D). The Freestyle Libre (black) and Dexcom (red) traces are shown alongside meal onset markers. Shaded regions indicate the incremental AUC (iAUC) computed above the pre-meal Libre baseline for each meal. Reference lines mark the sensor floor (40 mg/dL), normal fasting glucose (100 mg/dL), and the hyperglycemia threshold (180 mg/dL).

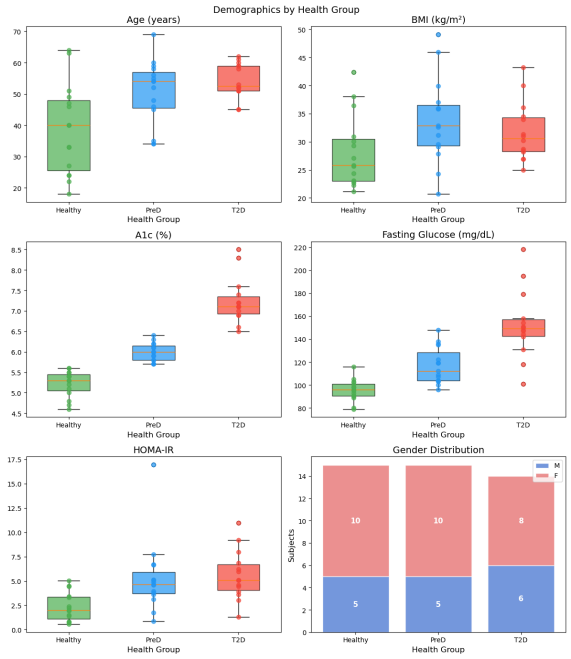



Figure 2: Distribution of key demographic and clinical features across the three metabolic groups

ing progressively greater insulin resistance across groups. Age and BMI show more overlap, suggesting they are weaker discriminators of metabolic status in this cohort.

Regarding the prediction targets, Figure 3 shows that iAUC values are right-skewed and highly variable across subjects (mean: 3132, std: 2884 mg/dL·min), consistent with the large inter-individual variability in postprandial glucose response reported in the literature (Zeevi et al., 2015). T2D subjects show the highest median iAUC across all meal types, while healthy subjects show the lowest, as expected from their metabolic profiles.

# 3 Methods

## 3.1 Experimental Design

Since each participant contributes with multiple meals across ten days, a random meal-level split would allow samples from the same individual to appear in both training and test sets, causing subject-level data leakage. To prevent this, all experiments were evaluated using **Leave-One-Subject-Out (LOSO) cross-validation**. In each fold, all meals from one participant were held out for testing, while the remaining participants were used for training. Features were standardized using a `StandardScaler` fitted on the training set of each fold and applied to the test set, ensuring no leakage through normalization. Final performance metrics were computed by aggregating predictions across all folds, following the evaluation protocol of the original CGMacros study (Das et al., 2025).

## 3.2 Linear Models: OLS and Mixed-Effects

As a linear baseline, we evaluated ordinary least squares regression (OLS) alongside three linear mixed-effects model (LMM) variants. Since each participant contributes with multiple meal observations, these measurements are not independent and may reflect subject-specific glucose response patterns. OLS treats all observations as independent and ignores within-subject correlation, while LMMs account for this dependency by including a random intercept per subject. Four configurations were evaluated:

- **OLS:** standard fixed-effects regression including meal type and health group as categorical covariates: iAUC $\sim$ features + Meal_Type + Health_group.
- **LMM-base:** extends OLS with a random intercept per subject to account for individual baseline differences in glucose response: iAUC $\sim$ features + Meal_Type + Health_group + $(1|\text{subject})$.
- **LMM-interaction:** extends LMM-base with a Meal_Type $\times$ Health_group interaction

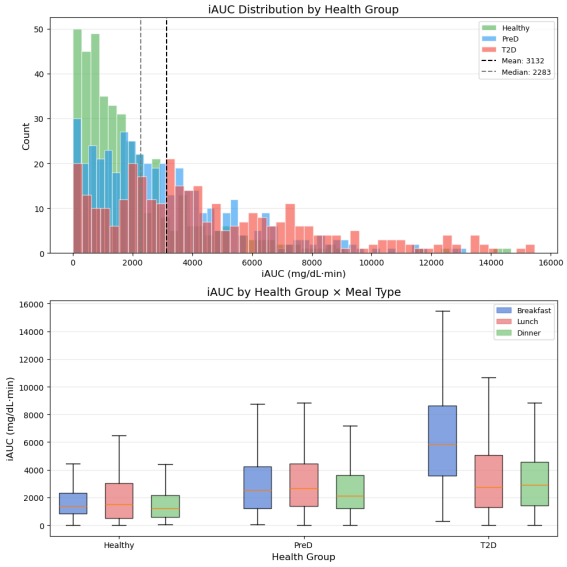



Figure 3: iAUC distribution plots

term, testing whether the effect of meal timing on glucose response differs across metabolic groups: iAUC $\sim$ features + Meal_Type $\times$ Health_group + (1|subject).

- **LMM-stratified:** three separate LMMs fitted independently for breakfast, lunch, and dinner, removing meal type as a covariate and instead modelling each meal type's glucose response with its own parameter estimates: iAUC $\sim$ features + Health_group + (1|subject). This configuration enables direct comparison with the meal-type stratified XGBoost models described in Section 3.3.

All models were fitted using `statsmodels` and evaluated under the LOSO cross-validation framework. Beyond predictive performance, the LMM-base was also fitted on the full dataset to obtain coefficient estimates and assess which features are statistically significant predictors of postprandial glucose response.

### 3.3 Non-linear Models: XGBoost

XGBoost was evaluated as a non-linear model. To enable comparison with the original CGMacros study, we adopted the same hyperparameter configuration: 80 estimators, learning rate of 0.2, decision stumps (`max_depth=1`), L1 regularization (`reg_alpha=1`), and no L2 regularization (`reg_lambda=0`). Depth-1 trees limit model complexity and help avoid overfitting given the small cohort size. Four configurations were evaluated:

- **Original:** baseline glucose, clinical variables, and meal macronutrients, trained on breakfast meals only.
- **Extended:** the original feature set augmented with pre- and post-meal physical activity variables (`Activity_pre_sum`, `Activity_post_sum`).
- **Stratified:** separate models trained independently for breakfast, lunch, and dinner using the extended feature set.
- **General:** a single model trained on all meal types using the extended feature set, with meal type included as an input feature.

The Original and Extended configurations were compared on breakfast meals to assess the contribution of physical activity features. The Stratified and General configurations were compared to assess the effect of meal-type stratification.

### 3.4 Foundation Model: TabPFN

We evaluated TabPFN (Hollmann et al., 2025), a transformer-based foundation model for tabular data pre-trained on a large collection of synthetic datasets. Unlike conventional supervised models, TabPFN performs inference through in-context learning, using the training set as context rather than fitting task-specific parameters. This makes it suitable for low-sample tabular settings such as the present study, where the cohort size is limited. TabPFN also includes internal preprocessing, such as feature normalization and missing-value handling, removing the need for external standardization.

TabPFN was evaluated with the same **General** and **Stratified** configurations as XGBoost, using the extended feature set and the same LOSO framework.

### 3.5 Evaluation Metrics

Model performance was assessed using four metrics: Pearson correlation coefficient ($r$), root mean squared error (RMSE), mean absolute error (MAE), and coefficient of determination ($R^2$). The primary metric was Pearson correlation, consistent with the original CGMacros study, as it measures how well model predictions follow the variability observed in glucose responses across individuals.

RMSE and MAE quantify prediction error in the original units (mg/dL·min). $R^2$ measures the proportion of variance in the target variable explained

by the model. All metrics were computed from predictions aggregated across all LOSO folds.

### 3.6 Explainability Methods

To interpret model predictions, we computed SHAP values (Lundberg and Lee, 2017) for both XGBoost and TabPFN. For each model, a single global model was trained on the full dataset to provide a population-level view of feature importance across all participants and meals.

For XGBoost, SHAP values were computed using `TreeExplainer`, which leverages the internal tree structure for efficient feature attribution. For TabPFN, which does not expose an interpretable internal structure, we used `PermutationExplainer`, which estimates feature contributions by measuring the effect of feature perturbations on model predictions. SHAP values were estimated using a background set of 100 samples and an explanation set of 300 samples.

For the linear mixed-effects models, we examined the fixed-effect coefficients directly as a complementary interpretability approach. These coefficients describe the average association between each predictor and glucose response at the population level, complementing the individual-level variability captured by the random effects.

## 4 Results

### 4.1 Linear Models

Table 2 summarizes the LOSO performance of all linear model variants for both iAUC and AUC prediction. Across all general models (OLS, LMM-base, LMM-interaction), iAUC prediction remained weak, with Pearson $r$ ranging from 0.327 to 0.365 and $R^2$ near zero or negative. AUC prediction was consistently stronger, with OLS achieving the best performance($r = 0.586$, $R^2 = 0.192$).

Meal-type stratification produced the most notable improvement for iAUC prediction. The breakfast-stratified LMM achieved $r = 0.496$ and $R^2 = 0.143$, the highest iAUC performance among all linear models, consistent with breakfast being the most standardized and predictable meal type in the dataset. Lunch showed moderate performance ($r = 0.422$), while dinner prediction effectively failed ($r = 0.135$, $R^2 = -0.321$), indicating that freely chosen, self-reported dinner meals are beyond the capacity of a linear model to predict reliably in a clinical setting.

Figure 4 summarizes Pearson $r$ across all linear model variants for both targets. Figure 5 shows predicted versus actual values for the stratified LMMs, illustrating the progressive deterioration in prediction quality from breakfast to dinner.

#### 4.1.1 Limitations

Notably, adding random intercepts (LMM-base) or interaction terms (LMM-interaction) did not improve over OLS under LOSO - and in some cases slightly degraded performance. This is an inherent limitation of evaluating mixed-effects models under a leave-one-subject-out framework: random intercepts are estimated during training for each of the 43 training subjects, but when predicting a completely new held-out individual, their random intercept is unknown and must be set to zero. The model therefore falls back to its fixed effects alone - essentially equivalent to OLS - but with fixed effect estimates that were calibrated assuming random effects would absorb part of the residual variance. This miscalibration can slightly hurt generalization compared to a plain OLS model that never modelled subject-level structure in the first place. A fairer evaluation of LMMs would involve fitting on the full dataset and examining coefficients and significance - which we will discuss further - rather than using them as predictive models for unseen individuals. Nevertheless, LOSO was retained as the evaluation framework for consistency across all model families compared in this work.

Table 2: LOSO performance of linear model variants.

| **Model** | $r$ | RMSE | MAE | $R^2$ |
|---|---|---|---|---|
| *iAUC* | | | | |
| OLS | 0.335 | 2859 | 2055 | 0.016 |
| LMM-base | 0.327 | 2915 | 2091 | -0.023 |
| LMM-interaction | 0.365 | 2884 | 2056 | -0.001 |
| **LMM-strat Breakfast** | **0.496** | **3181** | **2215** | **0.143** |
| LMM-strat Lunch | 0.422 | 2548 | 1884 | 0.088 |
| LMM-strat Dinner | 0.135 | 2518 | 1782 | -0.321 |
| *AUC* | | | | |
| OLS | 0.586 | 4500 | 2577 | 0.192 |
| LMM-base | 0.566 | 4565 | 2655 | 0.169 |
| LMM-interaction | 0.568 | 4561 | 2638 | 0.170 |
| **LMM-strat Breakfast** | **0.601** | **5328** | **2830** | **0.223** |
| LMM-strat Lunch | 0.569 | 3924 | 2407 | 0.126 |
| LMM-strat Dinner | 0.425 | 3962 | 2328 | -0.127 |

#### 4.1.2 Explainability

To gain interpretability beyond predictive performance, LMM-base was additionally fitted on the full dataset to examine which features are statistically significant predictors of iAUC. The results revealed four significant predictors ($p < 0.05$): **meal type** (Lunch: $\beta = -1083$, Dinner: $\beta = -1357$,

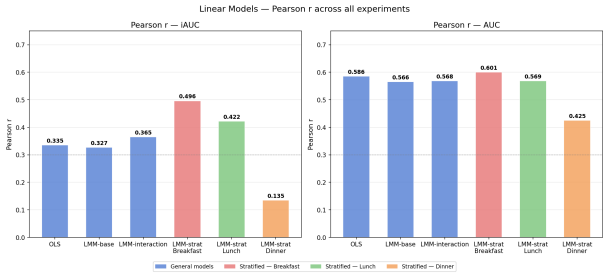



Figure 4: Pearson $r$ across all linear model variants for iAUC (left) and AUC (right). Blue bars represent general models, coloured bars represent meal-type stratified models.

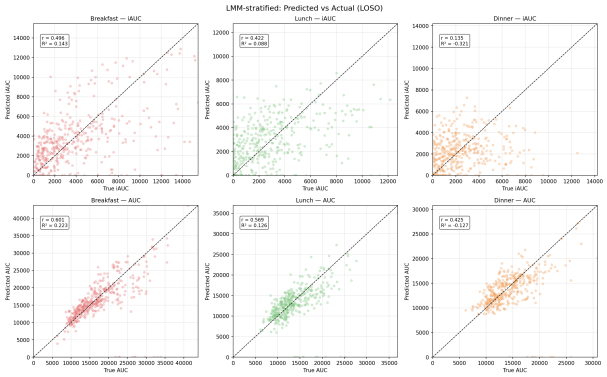



Figure 5: Predicted vs actual values for the LMM-stratified models under LOSO cross-validation. Rows correspond to iAUC (top) and AUC (bottom), columns to breakfast, lunch, and dinner. The dashed diagonal represents perfect prediction.

both $p < 0.001$), confirming that breakfast produces the highest glucose responses; **HbA1c** ($\beta = +1984$, $p = 0.003$), reflecting greater postprandial spikes with worse metabolic status; **baseline glucose** ($\beta = -28.2$, $p < 0.001$), a regression-to-the-mean effect where higher pre-meal glucose leaves less room for further rise; and **post-meal activity** ($\beta = -2.85$, $p < 0.001$), confirming that physical activity during the absorption window significantly attenuates the glucose spike. **Protein** content was also significant ($\beta = -4.4$, $p < 0.001$), consistent with the known glucose-attenuating effect of dietary protein.

## 4.2 Gradient Boosting (XGBoost)

### 4.2.1 Effect of Physical Activity

Table 3 reports LOSO performance on breakfast meals for the Original and Extended feature sets. The Original configuration showed similar performance to the results reported by Das et al. (2025), achieving $r = 0.642$ for iAUC and $r = 0.873$ for AUC, compared with $r = 0.645$ and $r = 0.885$ in the original study.

Adding pre- and post-meal physical activity resulted in a small improvement for iAUC prediction, increasing the Pearson correlation from 0.642 to 0.652 and $R^2$ from 0.411 to 0.424. In contrast, AUC performance remained almost unchanged, with only small differences across all metrics. These results suggest that activity features provide a limited but measurable contribution to predicting postprandial glucose response, while most of the predictive information for AUC comes from nutritional and clinical variables.

Table 3: LOSO performance of XGBoost on breakfast meals, original versus extended feature set ($n = 405$ meals, 44 subjects).

| **Configuration** | $r$ | RMSE | MAE | $R^2$ |
|---|---|---|---|---|
| *iAUC* | | | | |
| Original (19 features) | 0.642 | 2636 | 1890 | 0.411 |
| Extended (+activity) | 0.652 | 2608 | 1883 | 0.424 |
| Das et al. (2025) | 0.645 | — | — | — |
| *AUC* | | | | |
| Original (19 features) | 0.873 | 2980 | 2189 | 0.757 |
| Extended (+activity) | 0.875 | 2965 | 2176 | 0.759 |
| Das et al. (2025) | 0.885 | — | — | — |

Figure 6 shows predicted versus actual values for the Extended configuration. For iAUC, prediction errors increase at higher target values, particularly among T2D participants, who show the largest postprandial glucose responses. Healthy participants are mainly located at lower iAUC values, while prediabetic subjects show an intermediate pattern. In contrast, AUC predictions are more closely aligned with the identity line, consistent with the higher predictive performance observed for this target.

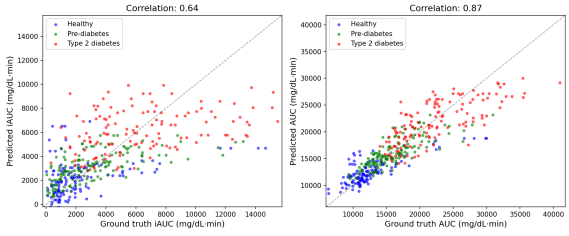



Figure 6: Predicted versus actual iAUC (left) and AUC (right) for the extended XGBoost configuration under LOSO cross-validation.

### 4.2.2 Meal-Type Stratification

Table 4 reports LOSO performance for separate meal-type models (Breakfast, Lunch, Dinner) alongside a single General model trained on all meals combined.

For AUC, all configurations achieved strong correlation, ranging from $r = 0.729$ (Dinner) to $r = 0.877$ (Breakfast). The General model ($r = 0.829$) performed competitively with the individual meal-type models, suggesting that overall glycemic exposure can be predicted reasonably well regardless of meal context.

iAUC prediction was considerably more variable across meal types. Breakfast yielded the strongest result ($r = 0.655$, $R^2 = 0.429$), consistent with its more predictable postprandial profile. This may reflect data quality and experimental control rather than a biological property of breakfast itself. Lunch performance dropped substantially ($r = 0.488$, $R^2 = 0.220$), and Dinner performance was notably weaker ($r = 0.257$, $R^2 = 0.018$), barely above chance. The General model ($r = 0.469$) fell between Breakfast and Lunch, reflecting the averaging effect across meal types. The difficulty in predicting dinner iAUC likely reflects the compounding variability accumulated over the course of the day - including effects of prior meals, cumulative activity, and circadian variation in insulin sensitivity - which the available features do not fully capture.

Table 4: LOSO performance of XGBoost stratified by meal type (extended feature set).

| **Configuration** | $r$ | RMSE | MAE | $R^2$ |
|---|---|---|---|---|
| *iAUC* | | | | |
| Breakfast ($n = 405$) | 0.655 | 2597 | 1877 | 0.429 |
| Lunch ($n = 369$) | 0.488 | 2355 | 1808 | 0.220 |
| Dinner ($n = 396$) | 0.257 | 2171 | 1567 | 0.018 |
| General ($n = 1170$) | 0.469 | 2550 | 1878 | 0.217 |
| *AUC* | | | | |
| Breakfast ($n = 405$) | 0.877 | 2943 | 2158 | 0.763 |
| Lunch ($n = 369$) | 0.794 | 2564 | 1945 | 0.627 |
| Dinner ($n = 396$) | 0.729 | 2557 | 1866 | 0.526 |
| General ($n = 1170$) | 0.829 | 2822 | 2072 | 0.682 |

### 4.2.3 Explainability

Figure 7 shows SHAP beeswarm plots for iAUC across all the stratified models. Fasting glucose and carbohydrate content are consistently among the top contributors, though their relative importance shifts by meal type. For breakfast, fasting glucose dominates - high values strongly increase predicted iAUC - while carbohydrates take the top position for lunch. For dinner, baseline glucose and HbA1c rank among the highest, with carbohydrates playing a comparatively smaller role, which may partly explain the weaker predictive performance for that meal type. Physical activity features (pre- and post-meal) appear consistently within the top ten across all stratified models, supporting the inclusion of activity as an informative signal.

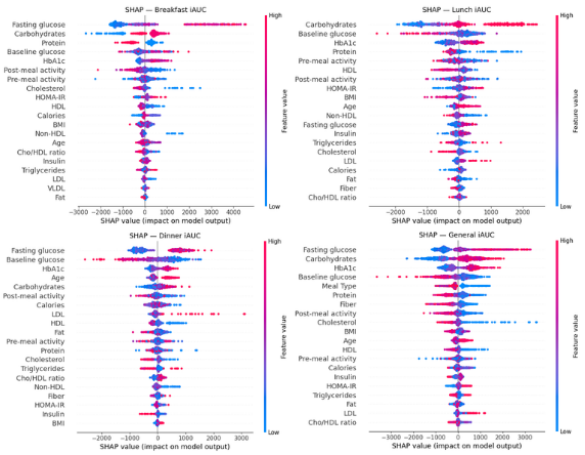



Figure 7: SHAP beeswarm plots for iAUC prediction across the four XGBoost configurations (Breakfast, Lunch, Dinner, General). Features are ranked by mean absolute SHAP value; colour indicates feature magnitude (red = high, blue = low).

For AUC (Figure 8), baseline glucose becomes the single most important feature across all meal types, which is expected, given that total glycemic exposure is strongly anchored to the pre-meal glucose level. HbA1c and fasting glucose follow consistently, reflecting the role of chronic glycemic control. Carbohydrates remain relevant but rank lower than for iAUC, suggesting that nutritional content has relatively more influence on the reactive spike than on the total glucose area. In the General model, meal type itself emerges as an important feature for both targets, confirming that postprandial responses differ systematically across breakfast, lunch, and dinner.

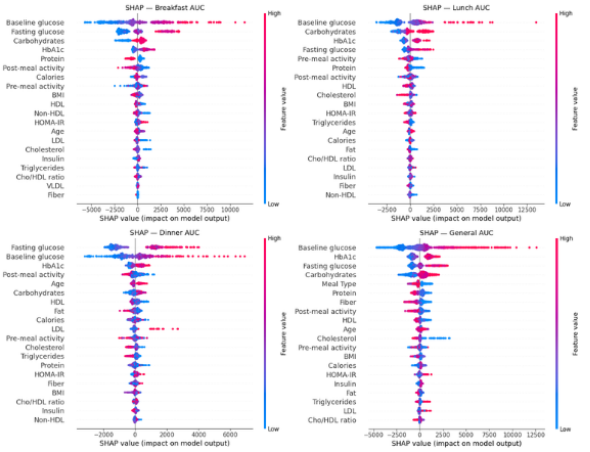



Figure 8: SHAP beeswarm plots for AUC prediction across the four XGBoost configurations (Breakfast, Lunch, Dinner, General). Features are ranked by mean absolute SHAP value; colour indicates feature magnitude (red = high, blue = low).

### 4.3 Tabular Foundation Model (TabPFN)

Table 5 summarizes the LOSO performance of TabPFN across all evaluated configurations.

For the stratified models, breakfast achieved the highest performance, with $r = 0.661$ and $R^2 = 0.430$ for iAUC. Performance decreased for lunch ($r = 0.529$, $R^2 = 0.256$) and further for dinner ($r = 0.336$, $R^2 = 0.060$), following the same trend observed for the other model families.

AUC prediction was consistently stronger than iAUC across all meal types. Even for dinner, TabPFN achieved $r = 0.786$ and $R^2 = 0.616$, indicating that AUC remains easier to predict than iAUC in this dataset.

For the general model, including meal type as an input feature improved iAUC performance from $r = 0.571$ to $r = 0.593$ ($\Delta R^2 = +0.027$). AUC showed a smaller but consistent improvement ($r = 0.860 \rightarrow 0.867$). These results suggest that meal context provides additional predictive signal even when a single model is trained across all meal types. However, both general configurations remained below the breakfast-specific model, indicating that separate models may better capture meal-specific patterns.

Compared with the Extended XGBoost configuration (Table 4), TabPFN achieved slightly higher performance on breakfast meals for both iAUC ($r = 0.661$ vs $0.655$) and AUC ($r = 0.892$ vs $0.877$). Across the other meal-type configurations, TabPFN also showed competitive performance, suggesting that the model can capture relevant patterns in this small tabular biomedical dataset. Since TabPFN requires no hyperparameter tuning and includes internal preprocessing, these results highlight its potential as an alternative to gradient boosting approaches in low-sample settings.

Table 5: LOSO performance of TabPFN across all evaluated configurations.

| **Configuration** | $r$ | RMSE | MAE | $R^2$ |
|---|---|---|---|---|
| *iAUC* | | | | |
| Breakfast | 0.661 | 2593 | 1871 | 0.430 |
| Lunch | 0.529 | 2301 | 1724 | 0.256 |
| Dinner | 0.336 | 2124 | 1545 | 0.060 |
| General | 0.571 | 2389 | 1727 | 0.313 |
| General+MT | 0.593 | 2341 | 1698 | 0.340 |
| *AUC* | | | | |
| Breakfast | 0.892 | 2795 | 2005 | 0.786 |
| Lunch | 0.796 | 2544 | 1915 | 0.633 |
| Dinner | 0.786 | 2301 | 1698 | 0.616 |
| General | 0.860 | 2634 | 1904 | 0.723 |
| General+MT | 0.867 | 2563 | 1858 | 0.738 |

Figure 9 shows predicted versus actual values for the General configuration. For iAUC, prediction errors increase at higher target values, and T2D points show opposite error tendencies between the two models: in XGBoost (Figure 6) they remain mostly below the identity line once true iAUC exceeds around 5000 mg/dL·min, reflecting underestimation of higher responses, while in TabPFN they sit mostly above it across the same range, reflecting a tendency toward overestimation for this subgroup. In contrast, AUC predictions are more closely aligned with the identity line for both models, consistent with the higher predictive performance observed for this target.

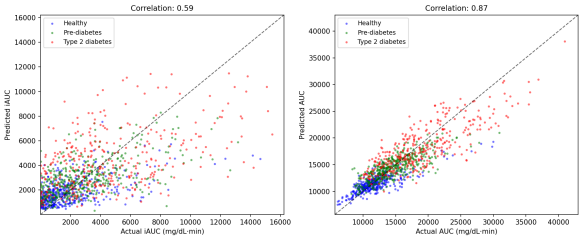



Figure 9: Predicted versus actual iAUC (left) and AUC (right) values for the TabPFN.

#### 4.3.1 Explainability

Figure 10 shows SHAP beeswarm plots for the TabPFN model trained on all meal types. For iAUC, carbohydrate content was the most influential feature, with higher values associated with higher predicted responses. HbA1c and fasting glucose were also among the top contributors, showing positive associations with predicted iAUC. Baseline glucose showed a negative contribution, indicating

that higher pre-meal glucose values were associated with smaller incremental responses after accounting for the other predictors. Meal type ranked among the important features, suggesting that it provides additional predictive information when all meals are modeled together. Post-meal activity was also among the relevant features, with higher activity values associated with lower predicted iAUC.

For AUC, baseline glucose was the dominant feature by a large margin, reflecting the direct contribution of the pre-meal glucose level to overall glucose response. This pattern was similar to the one observed for XGBoost.

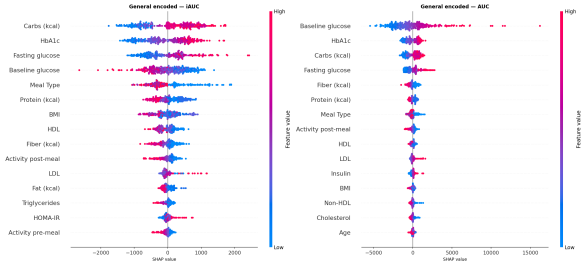



Figure 10: SHAP beeswarm plots for TabPFN trained on all meal types (iAUC left, AUC right).

For the breakfast-only model (Figure 11), fasting glucose ranked first for iAUC prediction instead of carbohydrates. This may be related to the standardized composition of breakfast meals, which reduces variation in nutritional features and increases the relative importance of individual metabolic characteristics. For AUC, baseline glucose again emerged as the most influential feature.

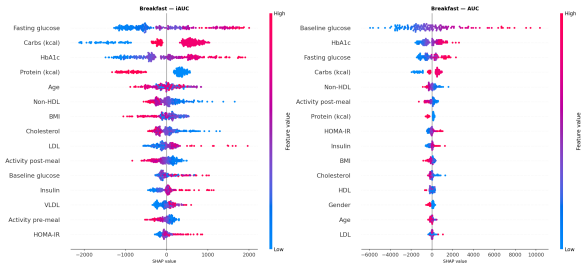



Figure 11: SHAP beeswarm plots for TabPFN trained on breakfast meals only (iAUC left, AUC right).

# 5 Discussion

## 5.1 Key Findings

This study compared linear models, gradient boosting, and a tabular foundation model for predicting postprandial glucose response from CGM, nutritional, clinical, and activity data.

The first finding was that meal context influenced prediction performance across all model families. Breakfast consistently achieved the highest accuracy, while dinner showed lower performance. This difference likely reflects the structure of the dataset, where breakfast meals were standardized, whereas dinner meals were self-selected and introduced greater variability.

Adding physical activity features provided additional information, particularly for iAUC prediction, although the improvement was limited. Across models, nutritional and metabolic variables such as carbohydrate intake, HbA1c, and baseline glucose remained among the most important predictors, highlighting the combined contribution of meal composition and individual metabolic characteristics.

TabPFN achieved comparable performance to XGBoost under the same LOSO evaluation framework, showing that a tabular foundation model can perform well in a small biomedical dataset without extensive model-specific tuning. The similar performance between models suggests that the quality and relevance of input features play an important role in prediction accuracy.

Feature importance analysis showed consistent patterns across models. Carbohydrate intake, HbA1c, fasting glucose, and baseline glucose repeatedly appeared among the most influential predictors, supporting the importance of both nutritional factors and individual metabolic state in explaining differences in postprandial glucose response.

## 5.2 Limitations

This study has several limitations. First, the dataset includes a relatively small cohort of 44 subjects. Although LOSO evaluation prevents subject-level data leakage, the limited number of participants may reduce the stability of performance estimates and should be considered when interpreting model comparisons.

Second, postprandial glucose response is influenced by biological factors that are not fully represented by the available features. The current models incorporate meal composition, clinical measurements, baseline glucose, and activity data, but additional information such as insulin dynamics and microbiome-related features was not explored.

Third, meal categories differ in terms of data consistency. Breakfast meals were standardized, whereas dinner meals were self-reported and more

variable. This difference in meal structure likely contributed to the lower prediction performance observed for dinner.

Finally, this study evaluates model performance within a single cohort and does not assess whether the models generalize to other populations. Validation on independent datasets will be necessary to determine whether the observed patterns extend beyond the current study population.

### 5.3 Future Directions

Future work could extend this study by incorporating CGM-based foundation models that operate directly on glucose time-series data. Such approaches may capture temporal information from glucose dynamics that is not represented when responses are summarized into meal-level features such as AUC and iAUC.

Another potential direction is the integration of additional biological information, including the gut microbiome features available in CGMacros. These variables may help explain individual differences in glucose response that are not fully captured by nutritional and clinical measurements alone.

## 6 Conclusion

In this work, we compared different machine learning approaches for predicting postprandial glucose response from CGM, nutritional, clinical, and activity data, including linear mixed-effects models, gradient boosting, and a tabular foundation model. The results show that postprandial glucose response can be predicted from available meal and metabolic information, although performance varies depending on the meal type and prediction target.

Linear mixed-effects models provided an interpretable baseline and highlighted the contribution of metabolic and meal-related factors, while nonlinear models achieved higher predictive performance. Meal-specific models performed better for standardized meals, particularly breakfast, whereas more variable meals such as dinner remained more difficult to predict. TabPFN achieved comparable performance to XGBoost across the evaluated configurations, showing that foundation models can extract useful information from small tabular biomedical datasets.

Based on these results, improving PPGR prediction requires not only selecting appropriate modeling approaches but also better capturing the variability between individuals. Larger cohorts and additional biological information, together with CGM time-series features, may help improve the generalizability of future models.

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